

5(a)(i)	$q^2 = 0.6$ or $q = \sqrt{0.6}$ or $q = 0.775$ or $p = 0.225$; $2pq = 2 \times 0.225 \times 0.775$ or 0.35 ; <i>allow ecf from mp1</i> 35 % ; <i>allow ecf from mp2</i>	3
5(a)(ii)	any four from: 1 no migration or no flow of, alleles / genes, into / out of, the population ; 2 population is isolated / described ; e.g. separated by geographical barrier I ref to island alone 3 random mating is occurring ; 4 rats move around island / AW ; 5 large population ; 6 rats have high reproductive rate / no predators ; 7 rats are diploid ; 8 have two different alleles (for the gene) / heterozygous ;	4

Question	Answer	Marks
5(b)	any three from: 1 new allele initially found in heterozygotes ; <i>dominant advantageous allele</i> 2 (new) dominant allele is, expressed in heterozygote / always expressed ; 3 selection can act on individuals with dominant allele straight away / AW ; <i>recessive advantageous allele</i> 4 (new) recessive allele is not expressed, in heterozygote / due to dominant allele or recessive allele only expressed in homozygous recessive (genotype) ; 5 selection can only act on individuals who are homozygous recessive ; 6 it takes time for homozygous recessive (genotype) to occur / AW ;	3